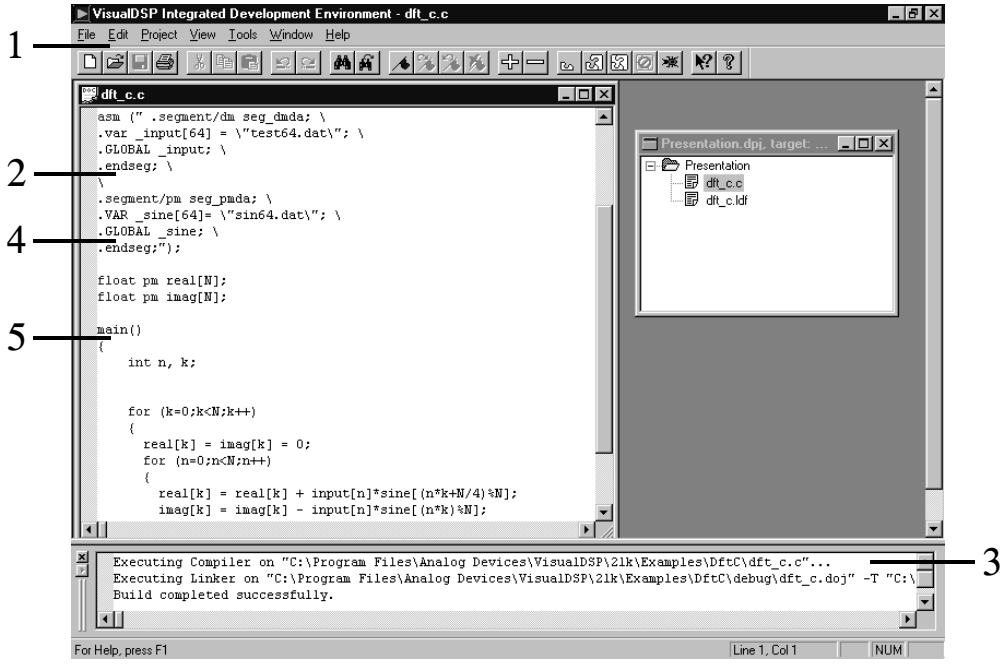

1. Move easily between Edit, Build, and Debug activities
2. Mix Assembly and C in a common source file
3. View "build" results
4. Powerful editor understands syntax
5. Easy access through bookmarks

FIGURE 29-7
Example screen from VisualDSP. This provides an integrated development environment for creating programs on the SHARC. All of the following functions can be accessed from this single interface: editing, compiling, assembling, linking, simulating, debugging, downloading, and PROM creation.

Table 29-4 shows an example C program, taken from the Analog Devices' "C Compiler Guide and Reference Manual." This program generates an echo by adding a delayed version of the signal to itself. The most recent 4000 samples from the input signal are stored in a circular buffer. As each sample is acquired, the circular buffer is updated, the newest sample is added to a scaled version of the oldest sample, and the resulting sample directed to the output.

The next advanced software tool you should look for is an **integrated development environment**. This is a fancy term that means everything needed to program and test the DSP is combined into one smoothly functioning package. Analog Devices provides an integrated development environment in a product called **VisualDSP®**, running under Windows® 95 and Windows NT™. Figure 29-7 shows an example of the main screen, providing a seamless way to edit, build, and debug programs.

Here are some of the key features of VisualDSP, showing why an integrated development environment is so important for fast software development. The editor is specialized for creating programs in C, assembly, and a mixture of the two. For instance, it understands the *syntax* of the languages, allowing it to display different types of statements in different colors. You can also